

Test

Definition 0.1. *dummy.*

Definition 0.2. *dummy.*

Theorem 0.1. *dummy.*

Proposition 0.1. *dummy.*

Proposition 0.2. *dummy.*

Proposition 0.3. *dummy.*

1 The Toolkit

2 The Calculus

3 Variants

4 Alphabet Invariance

Every construction so far has fixed its alphabet, and the two letters always sufficed. This section addresses the fourth open problem of Section 2 head on: *which functions are reachable over every alphabet with $|\Sigma| \geq 2$, and does a larger alphabet ever add power?* The answer is that, up to a canonical conjugation, the reachable class does not depend on the alphabet at all: for any two alphabets of at least two letters and any *good coding* between them, f is reachable over the one iff its conjugate is reachable over the other. The boundary of the invariance is exactly $|\Sigma| = 1$: the conjugation exists one way and cannot exist the other way, for a structural reason.

4.1 Comma-Free Codings

Definition 4.1. (Good Codings) *A coding is an injective monoid homomorphism $c : \Sigma^* \rightarrow \Gamma^*$ with $D_c \triangleq c(\Sigma)$ of uniform length $\ell \geq 1$ (a dictionary); write $c(S)$ for the letter-by-letter extension. The coding is good if D_c is comma-free (Golomb's condition; the theory of codes treats it under synchronizing codes $[?, ?]$): no $w \in D_c$ occurs in $u \cdot v$ at any of the junction-straddling positions $1, \dots, \ell - 1$ ($u, v \in D_c$).*

Comma-freeness is the Golomb analogue of the Comma Code Lemma for $\text{enc}_{b,x}^2$ (blocks xc): it is exactly the self-synchronization that makes the coding usable inside a scanning calculus. Without it the conjugacy class of a function depends on the coding, and the notion degenerates: a free assignment ℓ with $\ell(SS) = \ell(S)^2$, say, makes $X \mapsto XX$ conjugate to the unary squaring map. The good codings are the ones on which the conjugation is canonical.

Lemma 4.1. (Existence: the Double- a Family) *Let Γ have two distinct letters $a \neq b$ and $\ell \geq 3$. Then*

$$D_\ell \triangleq \{a a m b : m \in \Gamma^{\ell-3}, aa \not\leq m, m[0] \neq a\} \quad (1)$$

is comma-free, of uniform length ℓ ; over the binary alphabet $|D_\ell| = \text{Fib}(\ell - 2)$, so good codings exist between any two alphabets with at least two letters.

Proof. Every codeword starts with aa , contains aa only at position 0 (the interior m avoids aa and, for $\ell > 3$, does not start with a), and ends with b . In $u \cdot v$ the factor aa therefore occurs only at 0 and ℓ (the junction reads $b \cdot a$), and any occurrence of a codeword, starting as it does with aa , starts at 0 or ℓ – aligned. The count: m is b followed by an aa -free word of length $\ell - 4$, counted by $\text{Fib}(\ell - 2)$. (Verified: comma-freeness of all 21 dictionaries in the family, binary/ ternary/ quaternary, $\ell = 3..9$; sizes at $\ell = 3..12$ match $\text{Fib}(\ell - 2)$ with $\text{Fib}(1) = \text{Fib}(2) = 1$.) $\square$

Lemma 4.2. (Self-Synchronization) *Let c be good and $B \neq \epsilon$. The occurrences of $c(B)$ in $c(C)$ are exactly $\{\ell \cdot s : s \text{ an occurrence of } B \text{ in } C\}$; in particular every occurrence of a codeword inside a codeword-sequence starts at a multiple of ℓ .*

Proof. Uniform lengths first: an occurrence of $w \in D_c$ (length ℓ) lying wholly inside a single codeword (length ℓ) of the surrounding sequence starts at offset 0. Otherwise the occurrence straddles at least one junction $j\ell$ with $p < j\ell < p + \ell$, and its restrictions to the two flanking codewords exhibit w in their product at the junction-straddling position $p - (j-1)\ell \in [1, \ell - 1]$ – against comma-freeness. For the block $c(B)$: its first codeword $c(B[0]) \in D_c$ occurs at a multiple of ℓ by the first part, the block then covers whole codewords of $c(C)$, and these spell B ; the converse is immediate. (Verified: all $B \in \{a, b, c\}^{\leq 3} \setminus \{\epsilon\}$, $C \in \{a, b, c\}^{\leq 5} - 14,196$ cases, zero exceptions.) $\square$

Lemma 4.3. (Pass Transfer) *Let c be good. For all $A, C \in \Sigma^*$ and $B \in \Sigma^* \setminus \{\epsilon\}$,*

$$[c(A)/c(B)] c(C) = c([A/B]C), \quad (2)$$

and the same identity holds for the once, rightmost, and k -th occurrence primitives: the greedy occurrence lists correspond order-isomorphically under $s \mapsto \ell s$.

Proof. Induction along the scan. By Self-Synchronization the occurrence lists correspond, and the correspondence preserves order and disjointness (positions scale by ℓ), so leftmost, k -th and rightmost all transfer; it is direction-free, so the right-to-left variant transfers too. For the full pass, write $C = XBY$ with $|X|$ minimal: the first occurrence of $c(B)$ in $c(X)c(B)c(Y)$ is the aligned one at $\ell|X|$ (any earlier one would be an aligned occurrence inside $c(X)$, i.e. an occurrence of B in X), the scan resumes at $\ell(|X|+|B|)$ – aligned again, and the inserted $c(A)$ is never re-entered, matching the never-rescan clause of Definition 0.1. Both sides now split as $c([A/B]X) \cdot c(A) \cdot c([A/B]Y)$ with $|X|, |Y| < |C|$. (Verified: $A, C \in \{a, b, c\}^{\leq 3}$, $B \in \{a, b, c\}^{1..3} - 188,760$ leftmost cases; once, rightmost and k -th ($k \leq 3$) on the smaller box – 31,200 cases; zero mismatches.) $\square$

4.2 The Transfer Theorem

Definition 4.2. (Conjugate) For f an n -ary partial function on Σ^* and $c : \Sigma \rightarrow \Gamma^\ell$ a good coding, the conjugate $f^c : \text{image}(c)^n \rightarrow \text{image}(c)$ is $c \circ f \circ c^{-1}$ – a partial function in the sense of the restriction clause of Definition 0.2.

Theorem 4.1. (Direction 1) Let $E \in \text{Exp}_n$ over Σ and c good. Define E^c by: constants $W \mapsto c(W)$, variables and both constructors unchanged. Then

$$\llbracket E^c \rrbracket(c(\mathbf{S})) = c(\llbracket E \rrbracket(\mathbf{S})) \quad (3)$$

whenever the right-hand side is defined, and the two sides are undefined together (the pattern is ϵ on one side iff on the other). Moreover $|E^c| = |E|$ (node count; constant symbols grow ℓ -fold in length), $\deg(E^c) = \deg(E)$, and $\text{Safe}(E) \Rightarrow \text{Safe}(E^c)$; and if $\llbracket E \rrbracket$ is total then a total Γ -expression agreeing with E^c on the image exists – wrap the pattern once: $[R^c/\gamma(P^c)]E^c$ with $\gamma(X) \triangleq \text{if}(\text{eq}(X, \epsilon), b, X)$.

Proof. Induction on E . Variables and constants are immediate; concatenation commutes with c because c is a homomorphism. For the pass node the patterns and replacements are the c -images of the sub-values, and Pass Transfer applies at every value of the sub-expressions; the definedness clause is the ϵ -condition, and $c(B) = \epsilon$ iff $B = \epsilon$. Safety and degree are structural ($c(W)$ is a nonempty constant iff W is; $\deg$ reads the replacement, and the replacement's image has the same degree). (Verified: all 2,954 depth- ≤ 1 expressions over $\{a, b, c\}$ with constants ≤ 2 , on all 121 inputs of length ≤ 4 – 357,434 evaluations, 0 mismatches, 23,912 undefined on both sides; 48,000 further evaluations of random depth- ≤ 3 expressions; the same under once and rightmost semantics, 118,160 evaluations each; degree and safety over the full space, 2,954 expressions, 0 violations; node count equal in every case.) $\square$

Lemma 4.4. (Composition) If $c : \Sigma \rightarrow \Gamma^{\ell_c}$ and $d : \Gamma \rightarrow \Sigma^{\ell_d}$ are good, then $d \circ c$ is good: $\{d(w) : w \in D_c\}$ is comma-free over Σ , of uniform length $\ell_c \ell_d$.

Proof. An occurrence of $d(w)$ inside a sequence of d -images of D_c -codewords begins with $d(w[0]) \in D_d$, which by Self-Synchronization over D_d starts on the D_d -grid; the whole block $d(w)$ then covers D_d -codewords spelling $d(w[0]) \cdots d(w[\ell_c - 1])$, so by injectivity of d the word w occurs among the pre-image codewords, and if the block did not start on the outer $(\ell_c \ell_d)$ -grid, w would occur in a product of two D_c -codewords at a junction-straddling position – against the comma-freeness of D_c . (Verified: $\{d(c(\sigma))\}$ for $c : \{a, b, c\} \rightarrow \{x, y\}^6$ and $d : \{x, y\} \rightarrow \{a, b, c\}^4$, length-24 dictionary, comma-free.) $\square$

Lemma 4.5. (Retraction) Let $h = d \circ c : \Sigma \rightarrow \Sigma^{\ell_c \ell_d}$ with c, d good. Then h is good, and both h and a decoder H_h with $H_h \circ h = \text{id}$ are total L -reachable: h is $\text{rep}_{|\Sigma|}$ with the single characters as patterns and the h -codewords as replacements, H_h the same with the roles exchanged – rep_n places no restriction on its patterns (Theorem 0.1).

Proof. Comma-freeness is Lemma 4.4; on an h -image the codeword occurrences are aligned (Self-Synchronization), so the freezing semantics of rep_n tiles the text exactly, replacing each $h(\sigma)$ by σ . Both rep_n have constant, nonempty patterns, hence are total. (Verified: the h - and H_h -expressions against their intended semantics on all 40 inputs of length ≤ 3 over $\{a, b, c\}$; and as part of the roundtrip below.) $\square$

Theorem 4.2. (Transfer) *For all finite Σ, Γ with $|\Sigma|, |\Gamma| \geq 2$, every good coding c , and every partial function f on Σ^* :*

$$f \text{ is } L_\Sigma\text{-reachable} \iff f^c \text{ is } L_\Gamma\text{-reachable}, \quad (4)$$

and when f is total the Γ -witness may be taken total. The Γ -witness of Direction 2 is $H_h(E'_d(h(X_1), \dots, h(X_n)))$ with $E' = E^c$, $E'_d = (E^c)^d$, $h = d \circ c$; it has $\deg$ equal to $\deg(E)$ and node count $|E|$ plus a constant depending only on the two alphabets.

Proof. Direction 1 is Theorem 4.1. For Direction 2 let E' over Γ witness f^c . Every $h(X_i) = d(c(X_i))$ lies in $\text{image}(d)$, where E'_d computes d of the E' -value (Direction 1, for the coding d); so

$$\llbracket E'_d \rrbracket(h(\mathbf{X})) = d(\llbracket E' \rrbracket(c(\mathbf{X}))) = d(c(f(\mathbf{X}))) = h(f(\mathbf{X})),$$

and H_h returns $f(\mathbf{X})$ by the Retraction lemma. Definedness tracks f 's: h and H_h are total, and E'_d is defined at the h -images exactly when E is at $\mathbf{X}$. The degree and node claims follow from Theorem 4.1 and the constant patterns of the two rep_n . (Verified: the full roundtrip $E_0 \mapsto E' \mapsto E_{\text{final}}$ on all 155 depth- ≤ 1 expressions with constants ≤ 1 , on all 40 inputs of length ≤ 3 – 6,200 evaluations, 0 mismatches, 1,025 undefined on both sides; $\deg E_{\text{final}} = \deg E_0$ in all 155 cases, node overhead 528 for the (3, 2)-alphabet pair.) $\square$

Corollary 4.1. (Alphabet-Uniform Questions) *The right pass r , the once-hinge r_1 , and every k -th-occurrence node are coding-equivariant – they commute with every good coding, by Lemma 4.3 – so each of their reachability questions is a single question, not one per alphabet: reachable over one alphabet with at least two letters iff reachable over every. The same holds for rev by Theorem 4.3 below. Consequently every negative search of Sections 2–3 run over one alphabet excludes the conjugate over every alphabet of at least two letters.*

Proof. Equivariance plus Theorem 4.2: if F commutes with the good coding e , then a Σ -witness for F conjugates along c to a Γ -function which, composed with the retraction $q = c \circ e$, computes the native F over Γ – the identical construction as in Theorem 4.3, with the equivariance of Lemma 4.3 in place of the palindrome clause. (Verified end to end: the pass node over $\{a, b, c\}$ transferred to $\{x, y\}$ and retracted – a full binary expression in the leftmost calculus, 294 evaluations, 0 mismatches; once, rightmost and second occurrence as semantic compositions, 882 evaluations, 0 mismatches; equivariance of all four passes under the retraction codings, 2,352 cases, 0 mismatches; the doubling function as a further end-to-end witness, 31 inputs.) $\square$

4.3 Reversal

Reversal is the one equivariance that codings do not enjoy for free: a letter-by-letter reversal reverses *each codeword individually*, so $c(\text{rev } S) = \text{rev}(c(S))$ holds iff every codeword is a palindrome – a false claim in an earlier draft of this section, caught by the verification battery (60 failures on 31 inputs). The naive “rev on the code image” is not the reversal of the image; the repair is to retract along a coding whose outer half is palindromic.

Lemma 4.6. (Palindromic Dictionaries)

- (i) $c(\text{rev } S) = \text{rev}(c(S))$ for all S iff every codeword of c is a palindrome; then $e(\text{rev } T) = \text{rev}(e(T))$ for every such e .
- (ii) (Middle-marker family.) For $\mu \in \Sigma$ and $m \geq 1$, $\{\alpha \mu \text{rev}(\alpha) : \alpha \in (\Sigma \setminus \mu)^m\}$ is comma-free and palindromic, of size $(|\Sigma| - 1)^m$ – unbounded for $|\Sigma| \geq 3$.
- (iii) Over the binary alphabet the family degenerates (size 1), but size-3 palindromic comma-free dictionaries exist: $\{\text{aabaa}, \text{ababa}, \text{abbba}\}$ and $\{\text{aabaa}, \text{babab}, \text{bbabb}\}$.

Proof. (i) is immediate on one-letter words. (ii) μ occurs in each codeword only at the exact middle, and in $u \cdot v$ only at the two middles; an occurrence of a codeword has its μ at $p+m$, hence $p \in \{0, 2m+1\}$ – aligned. (iii) by inspection. (Verified: (ii) 20 dictionaries, ternary and quaternary, $m \leq 5$, all comma-free with the stated sizes; (iii) both dictionaries comma-free; exact maxima of palindromic comma-free sets over binary, by branch and bound: 1, 1, 3, 2, 7, 14 at $\ell = 3, 4, 5, 6, 7, 9$ – data, not a formula.) $\square$

Theorem 4.3. (Reversal Uniformity) $\text{rev} \in L$ over one alphabet with at least two letters iff over every: $\text{rev} \in L_\Sigma$ and a good $c : \Sigma \rightarrow \Gamma^\ell$, a good palindromic $e : \Gamma \rightarrow \Sigma^k$ (Lemma 4.6) and $q \triangleq c \circ e$ give

$$\text{rev}_\Gamma = H_q \circ \llbracket E' \rrbracket \circ q, \quad E' = E^c. \quad (5)$$

Proof. $q(T) = c(e(T))$, so $\llbracket E' \rrbracket(q(T)) = c(\text{rev } e(T))$ by Direction 1; the palindrome clause gives $\text{rev } e(T) = e(\text{rev } T)$, hence $\llbracket E' \rrbracket(q(T)) = c(e(\text{rev } T)) = q(\text{rev } T)$, and H_q returns $\text{rev } T$. The needed dictionaries exist for every pair of alphabets: from a source of at least three letters use the middle-marker family for e ; from a binary source, first move to a ternary alphabet along the size-3 binary dictionary of Lemma 4.6(iii) – a coding of three letters needs three codewords, and three exist – and then proceed as before; c itself is never required to be palindromic (Lemma 4.1 suffices), and $D_q = \{c(e(\gamma))\}$ is comma-free by Lemma 4.4. (Verified: $e = \{\mathbf{x} \mapsto \mathbf{bab}, \mathbf{y} \mapsto \mathbf{cac}\}$ over $\Sigma = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$, c the length-6 double- a coding to $\{\mathbf{x}, \mathbf{y}\}$; q and H_q as expressions, with the oracle $D = c \circ \text{rev} \circ c^{-1}$ standing in for $\llbracket E' \rrbracket$ (rev-reachability being the hypothesis): $H_q(D(q(T))) = \text{rev } T$ on all 63 inputs of length ≤ 5 , zero failures.) $\square$

Remark. (No Contradiction with the Native Results) Theorem 4.2 conjugates; Propositions 0.1 and 0.2 speak of *native* functions, and the two live in different conjugacy classes. The binary witness W of the leftmost deletion transfers to a ternary alphabet as the *block-level* function $[\epsilon/c(b)]_1$ on the image of c – it deletes an ℓ -block, not a character, and is not the native ternary $[\epsilon/\gamma]_1$; the negative ternary searches could not have seen the transferred witness in any case, its depth after transfer and retraction being far beyond the exhaustive depth-3 and randomized 4–7-pass spaces. Nor is the $\ell = 1$ coding a loop-hole: an injective character map is (trivially) good, and conjugating the binary once-deletion along it yields the native ternary function *restricted to a two-letter sub-alphabet* – reachable as a partial function, exactly the anchored-domain face of the phenomenon, and consistent with the resistance being about general position over three letters. Finally, the once-calculus – and the rightmost and k -th-occurrence calculi – have transfer theorems of their own: Lemma 4.3 is scan-direction- and multiplicity-agnostic, the first coded occurrence being the coding of the first native one. The marker-freshness obstruction of Proposition 0.2(iii) is a statement about native simulation within one alphabet and is untouched.

4.4 The Unary Edge

Proposition 4.1. (The Boundary $|\Sigma| = 1$)

- (i) $c(\mathbf{a}) = \mathbf{ab}$ is good ($\{w\}$ is comma-free iff w is primitive), and every unary-reachable function is the restriction of a binary-reachable one.
- (ii) There is no injective monoid homomorphism $\{\mathbf{a}, \mathbf{b}\}^* \rightarrow \{\mathbf{a}\}^*$: any h has $h(\mathbf{ab}) = h(\mathbf{a})h(\mathbf{b}) = h(\mathbf{b})h(\mathbf{a}) = h(\mathbf{ba})$ while $\mathbf{ab} \neq \mathbf{ba}$. So no coding-conjugation crosses the $|\Sigma| = 1$ boundary from above, and the transfer theorem is sharp: it holds for exactly the alphabets of at least two letters.
- (iii) Every unary L -expression computes a length map in the closure of $\{n \mapsto n, n \mapsto k\}$ under pointwise sum (concatenation) and $(g, r, p) \mapsto (n \mapsto r(n)[g(n)/p(n)] + g(n) \bmod p(n))$ (the pass; the constant case is $n \mapsto i[n/j] + n \bmod j$, and $[X_1/a]X_1$ squares). Every such map is eventually polynomially bounded, so $n \mapsto 2^n$ is unreachable at any size. The floor-halving $n \mapsto \lfloor n/2 \rfloor$ – the unary-domain function of Proposition 0.3 – is absent from every unary expression of size ≤ 7 (36,978 expressions, 2,091 distinct length maps on $n \leq 24$): every single pass leaks its residue $n \bmod j$, and subtraction is not among the closures.

Proof. (i) is Direction 1 for the coding c (verified: all 258 depth- ≤ 1 unary expressions on $a^0..a^{24}$, 6,450 evaluations, 0 mismatches, plus 12,000 deeper-random; alignment and the pass law, 566 cases). (ii) is the displayed computation. (iii): the pass law is the greedy scan of a unary string (verified: 20 (i, j) pairs); the growth bound is an induction – a pass outputs at most $|R| \cdot \lfloor |E|/|P| \rfloor \cdot |P| + |E| \leq |R||E| + |E|$ characters; the halving search is exhaustive at size ≤ 7 ($\lceil n/2 \rceil$ and $n \bmod 2$ are present at size 4, n^4 at size 10 via $\lfloor [X/a]X/a \rfloor ([X/a]X)$, verified). Of 480 total binary length profiles $n \mapsto |\llbracket E \rrbracket(a^n)|$ ($n \leq 12$), exactly two are realized

by no size- ≤ 7 unary expression nor any constant shift of one – bounded-search evidence for the strictness of (i)’s converse. $\square$

References